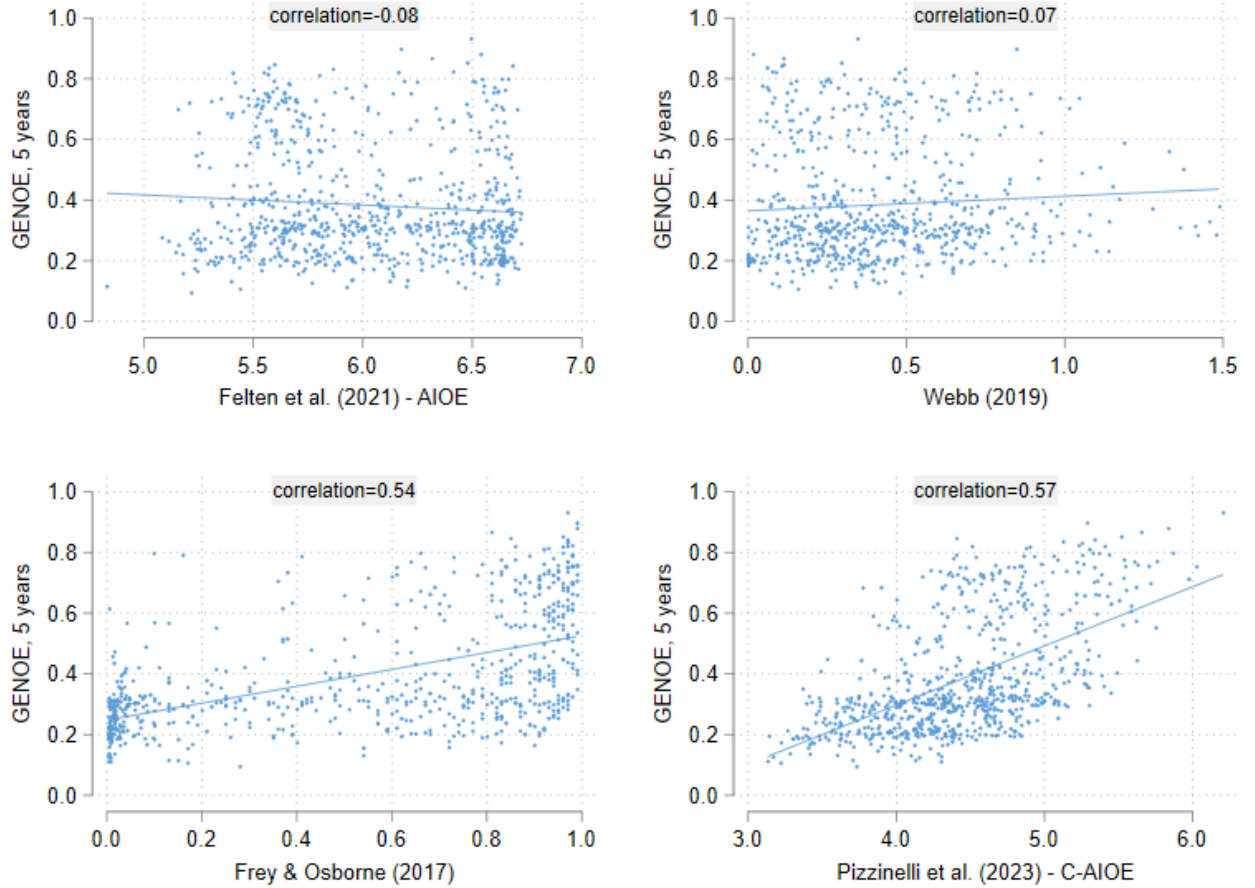


Figure 2: Correlation of GENOE and Other Indexes of Occupational Exposure



Source: Authors' formulation based on data from O*NET Felten *et al.* (2021); Webb (2019); Pizzinelli *et al.* (2023). Note: This figure displays the GENOE for the next one, five, and 10 years. The index measures the likelihood that occupations will be displaced by AI within each time horizon.

3.3 Alternative Methodologies

We explore how GENOE estimates vary under different methodologies and prompt choices. Our baseline specification involves providing the model with the occupation name and a vector of tasks and asking for an estimate of the likelihood of job replacement. We then introduce six alternative indexes, each employing a distinct approach:

Blind Index: This approach removes the occupation name, relying solely on the task vector for estimations. This allows us to isolate the impact of task-specific factors from occupation-specific context factors in the model's predictions and analyze if the absence of the occupation name significantly alters the estimates.

Task-by-Task Index: Instead of aggregating tasks under an occupation, we estimate the likeli-

hood of replacement for each task individually. These estimates are then averaged and weighted according to the relevance and importance of each task within the occupation, providing a granular view of task vulnerability.

Ethical Emphasis Index: Here, we explicitly ask the model to consider the ethical repercussions and potential social resistance to replacing certain tasks with AI. This approach highlights whether explicitly including ethical considerations causes a significant deviation from the baseline, which may already implicitly factor in such considerations.

Ethical Exclusion Index: This index explicitly asks the model not to consider ethical, regulatory, or social impediments to replacing tasks with AI. The objective of this index, compared to the previous one, is to measure the increase in the index when these considerations are excluded.

LLM-Specific Index: This variant focuses specifically on large language model (LLM) technologies. By narrowing the scope to LLM innovations, we assess the unique impact of advancements in natural language processing and related applications on occupational exposure, contrasting it with the general AI exposure measured in the baseline.

Non-AI Technologies Index: We estimate the probability of task replacement by non-AI technologies such as digital tools, robotics, and computer software. This provides a comparative baseline, distinguishing the impact of traditional technologies from AI-driven automation.

AI-Only Index: This index is derived by subtracting the Non-AI Technologies Index from the baseline GENOE, capturing the incremental exposure attributable solely to AI advancements. It allows us to identify occupations where AI-specific risks have significantly increased.

Table 4 provides the descriptive statistics of these alternative indexes, their correlation with the baseline GENOE, and other existing exposure indexes, and the extent to which they differ from the baseline. Both the Blind Index and the Ethical Emphasis Index do not deviate significantly from the baseline, although the mean differences are statistically significant. This indicates that there are slight changes in the reasoning of the model when the occupation name is omitted or when ethical considerations are explicitly prompted, but most of these factors are already incorporated in the baseline index. The overall correlations of these indexes with the baseline GENOE are 0.95 and 0.99, respectively. This consistency is supported by follow-up surveys, where ethical considerations and occupation-specific criteria were crucial elements for many of the model's assessments.

GENOE's ability to reflect ethical, regulatory, and social considerations in its assessment of AI's potential to replace human labor, even when not explicitly directed to do so, is a critical aspect of our argument. This aspect was initially confirmed by our follow-up survey analysis and is also supported by the comparisons with alternative indexes. Notably, when we explicitly instructed the model to include these factors, the resulting index showed only a minor reduction of one percentage point over the 5-year horizon. However, when we asked the model to exclude these considerations, the index increased by four percentage points, from 0.38 to 0.42. A similar pattern is observed for the 10-year horizon, as illustrated in Appendix Figure A.3, though the effect is less